Gowtham Reddy Peddireddy

B.Tech Student in Computer Science
Amrita Vishwa Vidyapeetham (2023–2027)
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Summary

Computer Science undergraduate at Amrita Vishwa Vidyapeetham with strong interest in Artificial Intelligence, Machine Learning, and Software Development. Skilled in Python, Java, Data Structures and Algorithms, and full-stack web development. Currently building AI-powered systems involving NLP, Retrieval-Augmented Generation (RAG), semantic evaluation, and real-time collaboration platforms. Passionate about solving practical problems using scalable and intelligent technologies.

Education

Amrita Vishwa Vidyapeetham, Coimbatore *2023 – 2027 (Expected)*
Bachelor of Technology in Computer Science
CGPA: 6.5

FIITJEE Junior College *2021 – 2023*
Intermediate Education with a focus on Science and Mathematics
Percentage: 92.4%

Coorg Public School, Gonikoppal *Completed 2021*
Secondary Education (CISCE Class X)
Percentage: 89.5%

Skills

- **Programming Languages:** Python, Java, C++
- **Web Development:** HTML, CSS, JavaScript, React, Flask
- **AI/ML:** NLP, RAG, Sentence Transformers
- **Libraries/Frameworks:** PyTorch, OpenCV
- **Core Concepts:** Data Structures, Algorithms, Computer Networks
- **Developer Tools:** Git, GitHub, VS Code

Projects

Evalify – AI-Powered Mock Interview Platform
Developing an AI-driven mock interview platform that evaluates candidate responses using NLP techniques and multiple assessment metrics including grammar, correctness, clarity, confidence, and communication quality. Implementing automated feedback generation, resume-based interview customization, and intelligent response analysis to simulate real interview experiences.

Equipoise – Biomedical Claim Verification using RAG
Built a Retrieval-Augmented Generation (RAG) system for biomedical scientific claim verification using the SciFact dataset. Compared multiple retrieval strategies including Dense Retrieval, BM25, Hybrid Search, and Query Reformulation to analyse how retrieval methods affect supporting and contradicting evidence discovery. Implemented semantic search, reranking, structured evidence synthesis, and evaluation metrics for balanced scientific reasoning.

Smart Industry Helmet Using STM32
Designed and developed a smart industry helmet using the STM32 platform to improve worker safety by detecting falls, monitoring hazardous gas levels and environmental conditions, and generating real-time alerts through buzzer, vibration, and Bluetooth communication.

Real-Time Collaborative Digital Canvas & Creative Workspace
Building a collaborative digital workspace supporting concurrent multi-user editing with low-latency synchronization, distributed state management, and real-time communication features.

Extracurricular Activities

- Coordinated Gokulashtami Event *2025*

Certifications

- Cybersecurity Analyst Job Simulation